

Markel Álvarez Alonso

Software Engineer & Cybersecurity

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PROFILE

Computer Engineer (UPNA) currently studying a Master's in Cybersecurity. Professional experience building production AI systems — LLM voice agents, RAG with OCR and vector search, AI-driven web automation — on Azure, Docker and GitHub Actions. Hands-on security projects in OT/ICS (IEC 62443, Suricata, MITRE ATT&CK) and DevSecOps.

EXPERIENCE

AI Software Engineer <i>Industrial Augmented Reality (iAR) — iAltech</i>	Jun 2026 – Sep 2026 Pamplona
- Hired after the internship to keep shipping production features across iAltech products: LLM agents, RAG pipelines and Azure cloud services. - Owned features end to end — implementation, integration testing, Docker deployment and CI/CD with GitHub Actions — working with real clients and deadlines.	
AI Developer (Intern) <i>Industrial Augmented Reality (iAR) — iAltech</i>	Feb 2026 – Jun 2026 Pamplona
- Designed conversational flows (prompts, state, conditional responses) for an autonomous phone AI platform and tuned NLP for noisy, colloquial calls (TypeScript, Azure Cognitive Services). - Extended a natural-language-to-SQL tool with PDF ingestion: image preprocessing, OCR, embeddings into a vector database and semantic search combined with SQL data (Python, Docker). - Built an AI-driven RPA with Playwright that breaks a goal into steps and navigates the web autonomously, with retries, UI-change detection and audit logs. - Developed a KPI analytics module for conversation logs (success rate, resolution time, drop-off points) with dashboards to guide product improvements. - Migrated an existing product to a new scalable architecture, with integration tests guaranteeing identical behaviour; reported QA issues in Azure DevOps.	

EDUCATION

Master's Degree in Cybersecurity <i>Universidad Pública de Navarra (UPNA)</i> Offensive and defensive security, network analysis and security engineering.	Sep 2026 – Present Pamplona
Bachelor's Degree in Computer Engineering <i>Universidad Pública de Navarra (UPNA)</i> Thesis: "The AI Paradigm Applied to Software Development: Research on AI-Based Methodologies and Tools to Enrich the Development Lifecycle". Includes a multimodal chatbot prototype (React, FastAPI, OpenAI, ElevenLabs).	2021 – 2026 Pamplona

PROJECTS

OT/ICS Security Lab <i>Python, Docker, Suricata, Modbus, MITRE ATT&CK, IEC 62443, nftables</i>	2025
- Virtual PLC (OpenPLC) controlling a simulated tank process - Controlled Modbus TCP attack mapped to MITRE ATT&CK for ICS	
DevSecOps Pipeline <i>GitHub Actions, DevSecOps, CI/CD, Python, CRA</i>	2025
- Security automated across the CI/CD delivery cycle - Each stage mapped to EU Cyber Resilience Act requirements	
todo-UNI <i>Next.js, React, Drizzle, Postgres, Gemini, Web Push</i>	2025
- Reads the inbox over IMAP looking for assignments and exams - Scrapes the university campus for new deadlines and notices	

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, Java, SQL, Bash
Frameworks: React, Next.js, Node.js, FastAPI, React Native, Expo, Tailwind CSS
Cloud & DevOps: Azure, Docker, Git, GitHub Actions, CI/CD, Linux, PostgreSQL, Supabase, Firebase, Azure DevOps
AI & Data: LLM, RAG, Embeddings, Vector DBs, OCR, NLP, OpenAI API, ElevenLabs, Playwright (RPA)
Security: DevSecOps, Suricata IDS, MITRE ATT&CK for ICS, IEC 62443, Modbus TCP, nftables, Wireshark, nmap

LANGUAGES

Spanish: Native | **Basque:** Native | **English:** Advanced — professional working proficiency (B2 certified)